

AdPilot OS

V2 — Training, Setup & Troubleshooting Guide

Meta & TikTok Paid-Ads Operating System

Read-only & safe by design — it analyses and **proposes**.
It never edits, pauses, or creates ads on your account.

Document version 2.0 · For agencies, freelancers & in-house teams
Live demo: edagher92-coder.github.io/SOCIALAUTOMATION · In-app manual: [/manual](#)

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The golden rule. AdPilot is **read-only**. It scores your campaigns and writes plain-English proposals. **You** decide and you make any change inside Meta/TikTok. Nothing here can spend money or alter a live ad.

Part 1 — Training

1.1 What AdPilot OS does

- AdPilot turns raw Meta & TikTok ad data into three things a busy operator actually needs:
- **A Campaign Health Score (0–100)** — one number that tells you if the account is healthy.
 - **Plain-English findings** — what's working, what's bleeding money, and why.
 - **Prioritised, safe recommendations** — keep / scale / reduce / refresh / fix-tracking / kill — always as proposals.
- It works for a single business or a whole agency: every client lives in its own isolated workspace.

1.2 Beginner vs Advanced mode

Mode	Who it's for	What you see
Beginner	Owners, new staff	Guided, plain-English view — the score, the headline issues, and the next best actions.
Advanced	Media buyers, analysts	Everything in Beginner plus the full 13-factor breakdown (“Why this score?”), per-factor weights and contributions.

Switch any time with the toggle in the top-left of the sidebar. Your choice is remembered.

1.3 The Health Score explained

The score is a weighted average of **13 factors**, mapped to a colour band:

Band	Score	What it means
Green	80–100	Healthy. Look for scale opportunities.
Yellow	60–79	Workable, but fix the flagged issues.
Orange	40–59	Underperforming. Act this week.
Red	0–39	Urgent. Money is likely being wasted.

The 13 factors

Tracking quality · CPA vs break-even · spend efficiency · conversion rate · click-through rate (CTR) · lead quality · creative freshness (fatigue) · cost per click (CPC) · naming convention · offer strength · landing-page alignment · budget pacing · data confidence.

Data confidence is the honesty factor — if there isn't enough spend/conversions to trust the result, AdPilot says so instead of pretending. In Advanced mode, open **“Why this score?”** to see each factor's score, weight, and points contributed.

1.4 Your daily / weekly workflow

1. **Sign in** → land on **Ads Health**.
2. **Get data in** — Sync connected accounts, or paste/upload a CSV.
3. **Confirm economics** in **Settings** (average sale value + gross margin → defines break-even).
4. **Analyse** → read the score, findings and proposals.
5. **Act in Meta/TikTok** on the proposals you agree with.
6. **Save / send** — open the report → **Download branded PDF** for the client.

1.5 Reading the results

- **Score hero** — the gauge, the band, and a one-line verdict.
- **Key metrics** — Spend, CPA, break-even CPA, ROAS, break-even ROAS, MER, leads, purchases.
- **Findings** — severity-tagged (Critical → Info) issues with a short explanation.
- **Proposals** — per-campaign verdict, the reason, and the exact suggested action.

1.6 Recommendations & safety

Verdict	Meaning
keep	Performing to target — leave it running.
scale	Strong & healthy — consider increasing budget (only suggested when health ≥ 70 and tracking is clean).
reduce	Slipping — trim budget / tighten targeting.
refresh	Creative fatigue — rotate in new creative.
fix-tracking	Conversions/pixel look broken — fix before judging performance.

kill	Wasting spend with no path to profit — pause it.
insufficient-data	Not enough signal yet — let it run / gather more.

Every verdict is a **proposal**. AdPilot never makes the change for you.

1.7 Creative library — link or upload

Bring AI audio, video and images from **any** tool — Midjourney, Sora, Runway, Pika, ElevenLabs, Canva, HeyGen, or your own files:

- **Link** — paste the asset URL (the universal method; works with every creator tool).
- **Upload** — store your own files in private storage.

AdPilot organises assets, links them to campaigns, and feeds them into creative briefs. *You bring the content — AdPilot does the rest.*

1.8 Multi-client (agency)

Use the **Client** selector (top-left) to switch workspaces, or **■ Add client...** to create one. Each client is fully isolated — data, connections, reports and branding never cross over (enforced by row-level security in the database).

1.9 Reports & one-click branded PDF

Every analysis auto-saves under **Saved Reports**. Open one → **Download branded PDF**. Set your agency logo, colour and support email under **White-label** so client reports carry *your* brand.

1.10 Notifications & alerts

Under **Notifications**: set an email, then enable the **weekly digest** and **critical alerts** (CPA blowouts, broken tracking, creative fatigue, Red health). A daily scheduled job scores connected accounts and emails any breaches. Use **Send test email** to confirm delivery.

1.11 AI prompting guide (Canva / Bobby / Aria)

Connect a Claude API key under **Claude API** to enable live AI. Then be specific:

Assistant	Give it
Canva Creator	Product + exact audience + the offer + the platform.

Bobby (Business)	A one-line description of your business + one concrete question.
Aria (Courses)	Topic + who it's for + the outcome a learner should reach.

AI assists — **you decide**. Always review AI output before using it.

Part 2 — Setup

2.1 Deployment essentials (owner / admin)

- AdPilot V2 runs on **Vercel** (the app) + **Supabase** (database, auth, storage). Minimum to go live:
1. Create a Supabase project; run the SQL migrations in `adpilot-v2/supabase/migrations` (through `0004_hardening.sql`).
 2. Deploy `adpilot-v2` to Vercel; set the environment variables (see §2.6).
 3. Set the two required keys: `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY`. Everything else is optional and degrades gracefully.

Before going live: rotate any previously-shared API keys (Anthropic, Stripe), set a real `SUPPORT_EMAIL`, and generate a fresh `TOKEN_ENCRYPTION_KEY` and `CRON_SECRET`.

2.2 Three ways to bring data in

Pick what suits you — you can mix them. **All paths are read-only.**

Method	Best for	Benefits	Drawbacks
CSV upload	Anyone, instant	Zero setup, no permissions, fully private, works offline	Manual; a point-in-time snapshot, not live
API token (OAuth)	Most users / auto-sync	One-click connect, scheduled auto-sync + scoring + alerts, least-privilege read scopes, tokens encrypted at rest	Requires a platform app configured once; tokens expire & refresh; app review for production
MCP server	Power users / Claude-driven	Drive it from Claude/agents, scriptable, great for multi-tool orchestration	More moving parts; you host the MCP; less turnkey

Recommendation: API token (OAuth) for daily auto-sync, CSV as the always-available fallback, MCP if you already orchestrate via Claude.

2.3 Option A — CSV (no setup)

Export at **ad level** with a daily breakdown from Meta Ads Manager / TikTok Ads Manager, then paste or upload on the **Ads Health** page. Columns auto-map to the universal schema (see Appendix B). Keep the header row.

2.4 Option B — API token (OAuth) — recommended

Server env keys (set once)

```
# Meta META_APP_ID=... META_APP_SECRET=... # TikTok TIKTOK_APP_ID=...  
TIKTOK_APP_SECRET=... # shared OAUTH_REDIRECT_BASE=https://your-app.vercel.app # no  
trailing slash TOKEN_ENCRYPTION_KEY=... # node -e  
"console.log(require('crypto').randomBytes(32).toString('base64'))"
```

Meta (Facebook / Instagram)

1. developers.facebook.com → **Create App** → type **Business** → add **Marketing API**.
2. App settings → Basic: copy **App ID + App secret** → `META_APP_ID` / `META_APP_SECRET`.
3. Facebook Login → Settings → **Valid OAuth redirect URIs**: add `https://your-app.vercel.app/api/oauth/meta/callback`.
4. Scopes: `ads_read`, `read_insights` (request in App Review for production; dev mode works for your own accounts).
5. In AdPilot: **Connect Accounts** → **Connect Meta** → approve → ad accounts appear → **Sync now**.

TikTok Ads

1. business-api.tiktok.com → **My Apps** → create an app; enable **Reporting / Ads (read)**.
2. Copy **App ID + Secret** → `TIKTOK_APP_ID` / `TIKTOK_APP_SECRET`.
3. Redirect URI: `https://your-app.vercel.app/api/oauth/tiktok/callback`.
4. Scope: `ads.read`.
5. In AdPilot: **Connect Accounts** → **Connect TikTok** → approve → **Sync now**.

Tokens are encrypted with **AES-256-GCM** before storage and are never sent to the browser.
Scheduled sync & scoring run via the daily cron once connected.

2.5 Option C — MCP server (Claude-driven)

If you run Meta/TikTok through an MCP server, have your agent export insights to the universal schema and push them to `POST /api/ingest` (authenticated with a session, or a server `x-api-key = INGEST_API_KEY` plus an `organisation_id`). It can also score and save in the same call. The schema is documented in `docs/MCP-INGEST.md`.

2.6 Environment-variable reference

Variable	Purpose	Required?
<code>NEXT_PUBLIC_SUPABASE_URL</code> <code>/..._ANON_KEY</code>	Database & auth	Yes
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Server-side admin writes	Yes (server)
<code>ANTHROPIC_API_KEY</code>	AI generation (Canva/Bobby/Aria)	Optional
<code>STRIPE_SECRET_KEY</code> / <code>STRIPE_WEBHOOK_SECRET</code>	Billing	Optional
<code>NEXT_PUBLIC_STRIPE_PRICE_STARTER</code> <code>/_PRO</code>	Plan price IDs (validated server-side)	Optional
<code>META_APP_ID/SECRET</code> , <code>TIKTOK_APP_ID/SECRET</code>	OAuth connect	Optional
<code>TOKEN_ENCRYPTION_KEY</code>	Encrypts platform tokens	If using OAuth
<code>CRON_SECRET</code>	Secures scheduled jobs (fail-closed if unset)	If using crons
<code>RESEND_API_KEY</code> / <code>EMAIL_FROM</code>	Email digests & alerts	Optional
<code>INGEST_API_KEY</code>	MCP/automation push to <code>/api/ingest</code>	Optional
<code>SUPPORT_EMAIL</code>	Shown in the manual / error reporting	Recommended

2.7 Post-connect checklist

- **Settings** — average sale value + gross margin per client.
- **Notifications** — weekly digest + critical alerts on.
- **White-label** — logo, colour, support email.
- **Crons** (in `vercel.json`) — daily auto-analysis + Friday digest, protected by `CRON_SECRET`.

Part 3 — Troubleshooting

3.1 Error reference

You see	It means	Do this
"No rows parsed"	CSV headers unrecognised	Export at ad level; keep the header row; use the universal template (Appendix B)
"AI generation isn't configured"	No Claude key	Set ANTHROPIC_API_KEY
"Billing isn't configured"	No Stripe keys	Set STRIPE_SECRET_KEY + price IDs
"No connected {platform} account"	Not connected	Connect Accounts → Connect → Sync now
"{platform}_not_configured"	App ID/secret missing	Set the META_/TIKTOK_APP_ID & SECRET
"Upload needs Supabase Storage"	Storage bucket not set up	Run migration 0003, or paste a link instead
"Could not analyse this data"	Engine rejected the rows (HTTP 422)	Check columns/values; try a sample CSV
401 / redirect to login	Session expired	Sign in again
Sync "502 ... API error"	Platform rejected the call	Token expired / rate-limited — reconnect or retry shortly
"Cron not configured"	CRON_SECRET unset (fail-closed)	Set CRON_SECRET in env

3.2 Diagnostics

- **Liveness:** [/api/health](#). Host status: Vercel dashboard.
- **Engine integrity:** a self-test runs on every build, confirming parity with the reference Python engine (30/30 checks).
- **Bad score?** Open "Why this score?", check **data confidence**, and verify the CSV columns.
- **Connection issues?** Re-check **Connect Accounts**; reconnect expired tokens.

3.3 Common issues & fixes

Symptom	Likely cause	Fix
Analyse does nothing	Empty CSV / bad headers	Confirm data & header row; try a built-in sample
Score looks wrong	Economics not set	Set avg sale value + margin in Settings; check data confidence
Seeing the wrong client's data	Active workspace	Switch with the Client selector (top-left)
PDF layout off	Print settings	Print dialog → A4/Letter → Save as PDF → enable “Background graphics”
No alert emails	Email not configured / rule off	Set <code>RESEND_API_KEY</code> + <code>EMAIL_FROM</code> ; enable the rule; Send test email

3.4 FAQ

- **Does it change my ads?** No — read-only, proposals only.
- **Is my data private?** CSVs are processed in-browser; connected data is org-isolated (RLS); tokens are encrypted.
- **Meta + TikTok together?** Yes — one schema, one combined score plus per-platform splits.
- **Do I need to code?** No.
- **Results guaranteed?** No — it surfaces issues and opportunities; performance depends on your execution.

3.5 How to report an error (so it gets fixed fast)

Email your support contact ([SUPPORT_EMAIL](#)) with:

1. What you did (the steps).
2. What you expected vs what actually happened.
3. The **exact error text** (screenshot is ideal).
4. Which client/workspace you were in.
5. A small sample of the CSV columns (no need for full data).

For developers: check Vercel & Supabase logs; the build self-test confirms engine integrity.

Appendix

A. Glossary

Term	Meaning
CPA	Cost per acquisition — $\text{spend} \div \text{conversions}$.
Break-even CPA	Average sale value $\times$ gross margin. Above it, you lose money per sale.
ROAS	Return on ad spend — $\text{revenue} \div \text{spend}$.
Break-even ROAS	$1 \div \text{gross margin}$.
MER	Marketing efficiency ratio — $\text{total revenue} \div \text{total ad spend}$.
CTR / CPC / CPM	Click-through rate / cost per click / cost per 1,000 impressions.
Creative fatigue	Performance decay as an audience sees the same creative too often.
RLS	Row-level security — database rules that isolate each client's data.

B. Universal CSV schema

Columns auto-map; these are the canonical names. Missing columns are handled — provide what you have.

```
business_name, platform, campaign_name, date, spend, impressions, reach, clicks,
leads, purchases, revenue, three_second_views, thurplays, lead_quality_score,
utm_source, utm_medium, utm_campaign, tracking_status
```

C. Break-even maths

- **Break-even CPA** = average sale value $\times$ gross margin
- **Break-even ROAS** = $1 \div \text{gross margin}$

Example: \$200 sale $\times$ 0.60 margin $\rightarrow$ break-even CPA \$120; break-even ROAS 1.67 $\times$.

D. Where things live

Area	Where
Score & analyse	Ads Health
Connect platforms	Connect Accounts
Saved reports / PDF	Saved Reports
Branding	White-label
Economics	Settings
Alerts & digest	Notifications
AI key	Claude API
This guide (in-app)	/manual

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